



SESI

SESI ARMADILLOS - PROJECT: SMILE TRIP

School SESI from Tatuí - SP // Brazil



Ana Hanf: Electrical Mechanics
EIGHTY



Davi Lucca: Programmer
SCIENTIST TEMPUS



Fernando Favier: Set Designer
TRIBOULET



Matheus Spesia: Mechanics
EDGY FROM EGYPT

PRESENTATION DESCRIPTION:

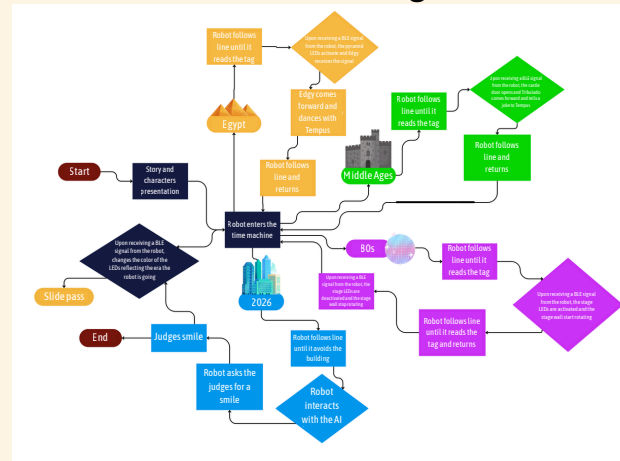
Chosen Features:



The narrative of our presentation follows the scientist Tempus in the year 3026, in a world where people no longer smile and society is devoid of joy. To reclaim the smiles the world has lost, the scientist, along with his assistant, ALL-IN, travels to the past in search of them. Throughout the presentation, he observes and shares the true reasons behind the smiles of the past, highlighting their importance for human connection and well-being. This entire performance is the result of a completely authentic and original creation developed by us, without inspiration from or reference to existing works.



Presentation overview diagram:



INSTAGRAM



OUR GITHUB



TECHNICAL VIDEO



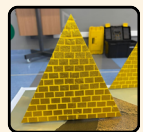
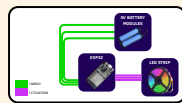
LOGBOOK

ELECTROMECHANICS:

OUR SCENERY:

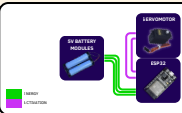
Pyramid

The Pyramid represents our Ancient Egypt setting. It features a WS2812B addressable LED strip controlled by an ESP32. The ESP32 waits for a signal indicating that ALL-IN has reached the Egypt era. Once this signal is received, the lighting effects and the Egyptian robot are activated, starting the scene and synchronizing the scenery with the robot's actions.

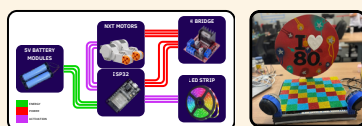


Castle

The Medieval Castle uses a similar system. When ALL-IN leaves the Time Machine and enters the Medieval era, it sends a signal to the castle's ESP32. The ESP32 then activates the motors that open the castle doors, allowing the era's character to emerge and begin the scene. This creates a synchronized transition between the robot, scenery, and story.



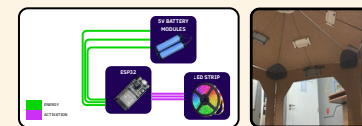
80s - System Architecture



The 1980s stage is designed to be colorful, dynamic, and fun. When ALL-IN leaves the Time Machine and enters this era, an ESP32 receives a BLE signal and activates the entire scene. The system turns on gyro-style lights, controls motors that rotate part of the stage background, and activates WS2812B addressable LED strips, creating a bright and energetic atmosphere inspired by the 1980s.

Time Machine

The Time Machine is one of the main elements of our story. It uses an ESP32 to control its LED system, changing the colors according to each era and the current moment of the performance. In addition, depending on the era, the Time Machine also changes the content displayed on the main screen, creating a synchronized connection between the physical scenery, lighting effects, and the story.



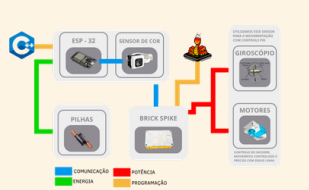
PROGRAMMING LANGUAGES:

For the programming architecture, we use different languages according to each system's requirements. Our LEGO SPIKE Prime robots are programmed in MicroPython, allowing them to communicate with the other devices through BLE. The smile-recognition AI was developed in Python, while the ESP32 controllers are programmed in C++. We also used JavaScript with the micro:bit during the development and testing process. In addition, VBA is used in PowerPoint to automate transitions and control the content displayed on the main screen during the presentation. For the LEGO EV3, we use MicroPython together with a micro:bit to enable BLE communication. Since the EV3 does not provide this functionality natively in our setup, the micro:bit acts as a communication bridge, allowing the EV3 to exchange BLE signals with the other systems in the project.

OUR ROBOTS:

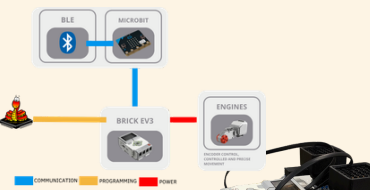
ALL-IN

Our ALL-IN robot uses a precise movement system based on line following with PID control, which helps stabilize its trajectory and improve movement accuracy. The robot is controlled by a LEGO SPIKE Prime running MicroPython. We chose this approach so that, even though ALL-IN is LEGO-based, it can still send and receive BLE signals and communicate with different microcontrollers and peripherals used throughout the project.



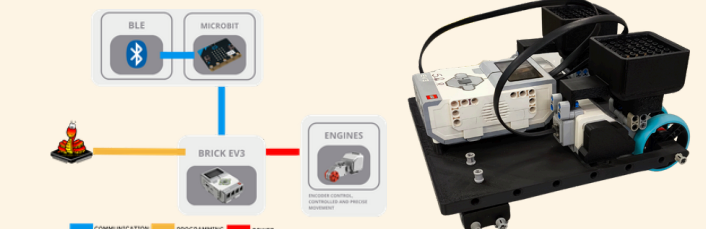
EDGY

The EDGY robot, representing the Ancient Egypt era, has a custom 3D-modeled base and is controlled by a LEGO EV3 running MicroPython. To enable BLE communication, the EV3 is connected to a micro:bit through serial communication. The micro:bit acts as a bridge, receiving and transmitting BLE signals for the EV3. This allows EDGY to remain fully integrated with the performance, being activated only at specific moments of the story.



TRIBOULET

The TRIBOULET robot, representing the Medieval era, uses the same communication architecture as EDGY. It is controlled by a LEGO EV3 running MicroPython, connected to a micro:bit through serial communication. The micro:bit acts as a BLE bridge, allowing TRIBOULET to send and receive signals from the other systems and activate only at the specific moments required by the story.

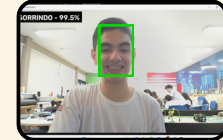
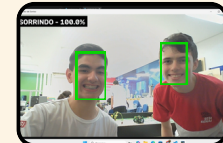
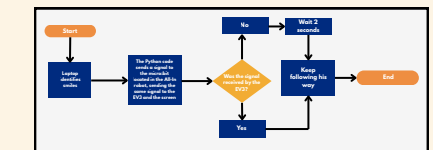


OUR FEATURES:



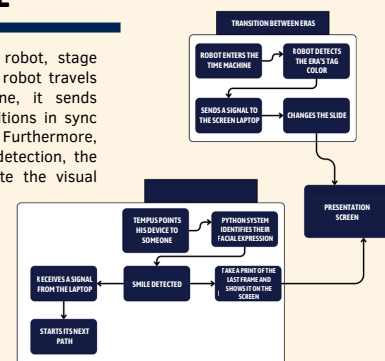
AI RECOGNITION

AI Recognition uses a custom YOLO model trained with around 4,000 labeled images to identify smiling and non-smiling expressions in real time. Running on a Lenovo 3-in-1 laptop with Python and OpenCV, the system analyzes the webcam feed, displays the detected expression and confidence level, and turns smile recognition into an interactive element of the presentation.



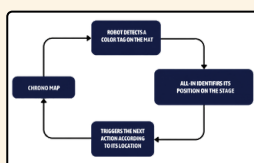
EXTENDED STAGE

This feature autonomously connects the robot, stage scenery, and main screen. Every time our robot travels through time eras via the time machine, it sends strategic signals so that our screen transitions in sync with the era the robot is currently in. Furthermore, during smile recognition moments, upon detection, the Python code also sends signals to update the visual assets.



CHRONO MAP

Chrono Map allows ALL-IN to understand where it is during the performance. The colored markings on the floor work as location tags, and the robot uses its sensors to identify each one. When ALL-IN detects a specific color, it recognizes its position on the stage and uses that information to understand the current part of the journey and decide what action should happen next.



COMMUNICATION ARCHITECTURE

Our presentation relies on Bluetooth Low Energy (BLE) broadcasting for real-time inter-device communication. By constantly transmitting low-latency BLE packets between the scenery and robots, the system coordinates automated movements, triggers live state transitions, and controls slide progression.

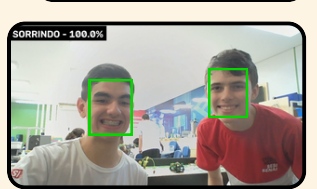
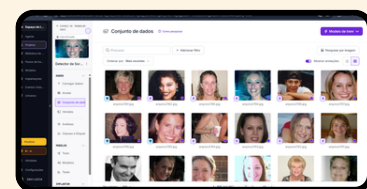
Bluetooth Low Energy



OUR INOVATIONS:

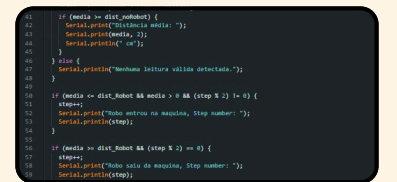
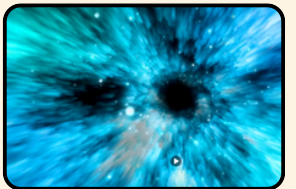
SMILE DEVICE

Our Smile AI was trained in Roboflow using a labeled dataset of smiling and non-smiling faces. We use YOLO for real-time recognition and Python to process the camera feed, display the detected expression, and integrate the result with the rest of the project.



USE OF THE SCREEN

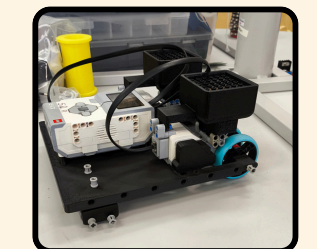
Our Extended Stage connects the physical scenery with the digital presentation. Through BLE signals, ESP32 controllers, and PowerPoint automation, the screen changes together with the robots, lights, and stage elements, creating one synchronized* experience between the real and digital parts of the story.



BLE COMMUNICATION

Our BLE communication architecture works like a broadcast network: each device can send signals that are received by all connected systems, while each robot or stage element only reacts to the commands intended for it. This allows the entire performance to stay synchronized without requiring direct one-to-one connections between every device.

	Voice	Data	Audio	Video	State
Bluetooth ACL / HS	x	x	x	x	x
Bluetooth SCO/HS	x	x	x	x	x
Bluetooth low energy	x	x	x	x	x
Wi-Fi (WIFI)	x	x	x	x	x
Wi-Fi Direct	x	x	x	x	x
Zigbee	x	x	x	x	x



EV3DEV AND MICROCONTROLLERS

Our EV3 communication innovation combines EV3Dev/MicroPython, serial communication, and BLE. Because the EV3 does not handle our BLE system directly, a micro:bit works as a bridge, exchanging data with the EV3 through serial while sending and receiving BLE signals from the rest of the project. This allows older EV3 robots to become part of the same modern wireless architecture.

